VCRSoft Consultation Transcript

Company: Mercer Heating & Air

Business type: HVAC Contractor

Engagement: 1

VCRSoft

Hi Dan Mercer — I'm an AI consultant that can help you leverage AI tools and solutions to grow Mercer Heating & Air. Let's start with the shape of your HVAC business: what share of your work is service/repair, system replacement/installation, maintenance agreements, and commercial or other work; who are your main customer types; and how many field technicians or crews do you currently run?

YOU — Turn 1

We're mostly residential here in Austin. About 45% of our revenue is service and repair, around 35% is replacement and installation, about 15% is maintenance agreements, and the last 5% is commercial and other work. We've got roughly 4,600 active residential customers — about 2,900 of them have called us in the last two years — plus 14 small commercial accounts. On the field side I run 9 service technicians and 4 installers, working as two crews of two. We cover the Austin metro plus Round Rock, Cedar Park and Pflugerville out of our shop on Burnet Road.

VCRSoft

Operating out of Burnet Road with 9 service techs and 4 installers across 4,600 active residential accounts gives us a clear view of your physical footprint and field capacity. Roughly what are your annual revenue and your typical monthly volume of service visits or completed jobs?

YOU — Turn 2

We're doing about $2.4 million a year in revenue now — $2.15 million last year and $1.9 million the year before that, so we're growing roughly 12% a year. On the service side we run about 310 repair jobs a month on average, and around $200,000 a month in revenue overall.

VCRSoft

Growing from $1.9 million to $2.4 million over three years while maintaining 310 monthly service repairs across your active residential base shows steady momentum. What is your primary growth objective, what measurable target would represent success, and by when?

YOU — Turn 3

I want to grow from $2.4 million to $3.1 million in revenue over the next 12 months, and lift our net margin from 7% to 12% — that's about $372,000 a year in net profit. That's the target I'd call success.

VCRSoft

Hitting $3.1 million in revenue and $372,000 in net profit requires bridging a $700,000 revenue gap while expanding net margin by 5 percentage points. What monthly operating budget and one-time setup budget would you consider for an improvement if the expected return is clear?

YOU — Turn 4

I'd go $3,000 to $5,000 a month in operating cost, plus up to $10,000 one-time to set it up — as long as it pays back within 6 months. We've got about $145,000 in cash, so that's comfortable.

VCRSoft

With $145,000 in cash and a 6-month payback requirement, any operational investment to bridge your $700,000 revenue gap has to be tightly tied to converting existing inquiries into completed jobs. For your emergency service calls, planned repairs, and replacement inquiries, how does an inquiry typically come in, and what steps does it go through from that first contact to a completed and paid job?

YOU — Turn 5

Most inquiries come in by phone or online — Google and search bring about 30%, referrals 28%, repeat customers 22%, aggregators 12%, and the rest another 8%. A CSR takes the call, books the visit — on average it takes 1.8 days from the call to a booked service visit, same-day for emergencies, and estimates get booked within 2 days. Our first response runs about 22 minutes normally but stretches to 47 minutes in summer. The dispatcher then puts it on the board in ServiceTitan by hand and the tech goes out. Residential customers pay at completion, 92% by card or check and 8% on financing; commercial is net-30.

VCRSoft

Taking up to 47 minutes for a first response during summer peak while leads arrive across search and referrals creates a risk of leakage before a visit is ever booked. Once a service visit or an estimate is booked and the technician goes out, what percentage of those visits convert into approved repairs or system replacements?

YOU — Turn 6

Overall, about 38% of leads turn into a job, and about 46% of leads get booked for an estimate. On the replacement side, our comfort advisor closes about 41% of estimates — 48% in summer, 33% in the shoulder months. On repairs, it's thinner: about 26 repair visits a month are genuine replacement opportunities, and only about 18% of those convert to a replacement within 90 days, about 5 a month.

VCRSoft

Across your 9 service technicians and 4 installers working out of Burnet Road, how do you manage daily dispatch utilization, drive times across the Austin metro, and parts availability, and which of those operational factors most limits your daily job throughput?

YOU — Turn 7

It's all manual — one dispatcher, a whiteboard and the ServiceTitan board, scheduled by phone and text. We run 4.4 jobs per tech per day in summer and 2.6 in the shoulder months. Drive time runs about 1.4 hours per tech per day because jobs get booked in call order, not by zone. First-time fix is 72% against an 85% target, and about 12% of jobs — around 37 a month — need a callback visit: 32% because the part wasn't on the truck, 28% misdiagnosis, 40% waiting on a part order. Truck stock is only loosely standardised, and emergency parts runs cost us about $9,000 a month. Honestly the manual dispatch and routing is what limits throughput most — it burns drive time and the callbacks eat slots we could be selling.

VCRSoft

A 1.4-hour daily drive time and 37 monthly callbacks driven by unstandardized truck stock and manual dispatching directly constrain the capacity you need to bridge your $700,000 revenue gap. How do you keep technician capacity and cash flow balanced between your peak heating and cooling periods and your shoulder seasons?

YOU — Turn 8

Honestly, we don't balance it well — we swing between the two extremes. In summer, June through August, we're at about $290,000 a month in revenue, tech utilization runs 95% plus, the backlog for non-emergency work stretches 2 to 3 weeks, and overtime runs about 14 hours per tech per week in July, which costs us around $18,000 a month in overtime premium. We lose about 22 booked jobs a month to competitors while customers wait, and the CSRs field around 90 'when can you come?' calls a week. Then in the shoulder months, March-April and October-November, revenue drops to about $130,000 a month, utilization falls to

about 55%, and the techs are idle about a day and a half a week each — that's around $24,000 a month in idle labour across the crew. Winter runs about 72% and $165,000 a month. My instinct has been that hiring two more technicians is the obvious solution, at about $68,000 loaded each, but I haven't worked out what two more techs would cost me sitting idle in the shoulders.

VCRSoft

Swinging between 95% utilization with $18,000 a month in summer overtime and 55% utilization with $24,000 a month in idle shoulder-season labor shows that adding two technicians would compound your seasonal imbalance rather than solve your revenue gap. What are your current operating margins and gross profit across your service, replacement, and maintenance revenue streams?

YOU — Turn 9

Our net margin is 7% now, about $168,000 a year or $14,000 a month — down from 9% last year and 12% two years ago, even as revenue grew. On gross margin, service and repair runs about 58%, and replacement is about 32% quoted, though we actually realise closer to 26% once ductwork surprises, permit delays and change orders land. On the maintenance agreements, I don't track the cost per agreement visit or the margin on the plan — that's a gap in how I run the business.

VCRSoft

Realising 26% on replacements against a 32% quote due to ductwork surprises, permit delays, and change orders eats into the margin needed to bridge your $700,000 revenue gap, especially when combined with unmanaged maintenance agreement costs. How does customer demand compare with your available capacity during your peak summer months versus your shoulder seasons, and how do you manage backlogs or fill unused availability across those periods?

YOU — Turn 10

In summer, demand runs well ahead of capacity — about 240 leads a month, around 460 service jobs a month, and 22 replacements a month, against a 2 to 3 week backlog for non-emergency work. We don't really manage the backlog; we work the overtime and lose about 22 booked jobs a month to competitors while customers wait. In the shoulders it's the opposite: about 140 leads a month, around 190 service jobs a month and 7 replacements a month. The maintenance agreements help fill some of that — the tune-ups for our 1,850 agreement customers mostly land in spring and fall — but we're still only at 55% utilization with about $24,000 a month in idle labour. We don't have a real plan for shifting work between the seasons.

VCRSoft

With 1,850 maintenance agreement tune-ups concentrated across your spring and fall shoulder months, your 55% utilization and $24,000 monthly idle labor cost point directly to how those visits are scheduled. What specific portion of your 1,850 maintenance agreement tune-ups are strictly restricted to spring and fall shoulder months versus being eligible for flexible scheduling into winter or early summer?

YOU — Turn 11

None of them are strictly restricted to spring and fall by the agreement itself — the plan promises two tune-ups a year, priority scheduling and a 10% repair discount, not specific months. They land in the shoulders because that's how we've always scheduled them, by phone, season by season. As far as I'm concerned all 1,850 agreements — 3,700 tune-up visits a year — are eligible to be scheduled more flexibly, including into winter; we've just never done it that way. And it's worth saying each of those visits costs me about $97 against $94.50 in revenue per visit, so the visits themselves don't make money before the repair discount.

VCRSoft

Operating tune-ups at a slight loss on every visit makes your 1,850 maintenance agreements a net drag during shoulder months when scheduling is concentrated by hand. What percentage of your 37 monthly

callbacks and emergency parts runs ($9,000/month) involve standard residential repair parts that could physically fit on standardized truck stock?

YOU — Turn 12

I haven't split it that way. What I know is about 32% of our 37 monthly callbacks — roughly 12 a month — happen because the part wasn't on the truck, and another 40% are waiting on a part order, with the other 28% down to misdiagnosis. The emergency parts runs cost us about $9,000 a month. I don't track what share of those are standard residential parts that would physically fit on a standardised truck versus parts that have to be ordered in.

VCRSoft

Losing 6% of replacement gross margin between the quote and final realization across 35% of your revenue creates a steady profit leak that directly undermines your goal of lifting net margin from 7% to 12%. What is the average dollar cost overrun per replacement installation caused specifically by ductwork surprises versus permit delays or change orders?

YOU — Turn 13

Ductwork surprises are the big one — about $850 a job on 30% of installs, and around 6 installs a month run over quote. Change orders run about $640 on average and hit about 18% of installs, and around 40% of them don't get billed until after the job is closed. Permits I can't put a dollar overrun on: every replacement needs an Austin permit at about $185 each and 3 to 5 days, and they delay installs, but I haven't priced what the delay itself costs us.

VCRSoft

Taking a $5,100 monthly hit on ductwork surprises across 6 installations while burning $9,000 in emergency parts runs points directly to estimating and truck inventory gaps that compound your margin pressure. What percentage of your 37 monthly callbacks and emergency parts runs involve standard residential repair parts that could physically fit on standardized truck stock?

YOU — Turn 14

The number I can stand behind is the 32% — about 12 of the 37 callbacks a month happen because the part wasn't on the truck. Those are the ones standardised truck stock would have covered. The other 40% are callbacks waiting on a part order, parts we have to order in, and 28% are misdiagnosis. I don't have a separate measured share for the $9,000 a month in emergency parts runs, but the callbacks tell the same story: the part not being on the truck is a third of the problem.

VCRSoft

A third of your callbacks being driven by parts missing from the truck directly compounds the 1.4-hour daily drive time from manual dispatching, eating the billable capacity you need to capture summer demand instead of losing 22 jobs a month to competitors. With 12 monthly callbacks caused by missing truck parts and unstandardized vehicle inventory across your 9 service vans, how much of your $9,000 monthly emergency parts spend is driven by technicians running out to local supply houses for common items during active jobs?

YOU — Turn 15

The $9,000 a month is the emergency parts runs themselves — trips to the supply house when a part isn't on the truck, usually mid-job. I don't split that spend between common items and special-order parts. What I can say is the callbacks show about 12 a month are parts not on the truck, so a good share of those runs are for parts that should have been on the van.

VCRSoft

With 12 monthly callbacks caused by missing truck parts and unstandardized vehicle inventory across your 9 service vans, a substantial portion of your $9,000 monthly emergency parts spend is driven by technicians running out to local supply houses during active jobs. What percentage of those 12 monthly truck-stock callbacks and emergency parts runs involve standard residential repair parts that could physically fit on a standardized van inventory?

YOU — Turn 16

It's the 32% — about 12 of the 37 callbacks a month happen because the part wasn't on the truck. Those are standard residential repair parts: the tech fetches them from the supply house the same day, so they are exactly the parts a standardised van should carry. The other 40% of callbacks are waiting on a part order — parts that have to be ordered in and wouldn't be on the van.

VCRSoft

With 12 monthly callbacks caused by missing truck parts and 40% of callbacks waiting on special orders, your $9,000 monthly emergency parts spend and 1.4-hour daily drive time are symptoms of unstandardized truck stock and manual dispatching directly compounding your summer capacity crunch. WORKING DIAGNOSIS: Mercer Heating & Air is constrained by seasonal capacity bottlenecks in summer and severe idle labor in shoulder months, meaning that driving more unmanaged demand without route optimization and seasonal work-shifting will only increase overtime waste and lost summer jobs. Does that match your experience, and what am I missing?

YOU — Turn 17

Yes, that matches my experience — summer is the bottleneck and the shoulders are dead time, and more unmanaged demand would just mean more overtime and more lost jobs. What you're missing: first, the replacement side is leaking. About 26 repair visits a month are genuine replacement opportunities and only about 18% convert within 90 days — my techs aren't trained or incentivised to present replacement, and the comfort advisor only sees the customer if the tech flags it, about 55% of the time. Second, we've got about $310,000 in open estimates and around 40% go stale because follow-up is manual and stops after 7 days. Third, I'm spending $5,500 a month on Google Ads and I don't know what our Google Ads spend actually returns in signed replacements. And hiring isn't simple either — I've had one service-tech role open 5 weeks with 2 candidates, neither of them EPA-608 certified, and I'm still working about 55 hours a week myself, 25 of that in the field and dispatch.

VCRSoft

Uncovering 26 repair opportunities a month that leak because comfort advisors only get called 55% of the time, alongside $310,000 in open estimates going stale after 7 days and $5,500 monthly on unmeasured Google Ads, shows that demand capture and conversion are leaking revenue just as much as seasonal capacity. How do you currently track and manage those open estimates once they pass the initial 7-day manual follow-up window?

YOU — Turn 18

We don't, really — it's a manual paper list. The comfort advisor works through about 55 estimates a month, and after about 7 days with no follow-up they just go stale — around 40% of them, out of about $310,000 sitting open. There's no automated follow-up and nothing that flags them after the first week.

VCRSoft

A manual paper list that drops open estimates after 7 days leaves a large share of your $310,000 pipeline unworked. How many total estimates are sitting in that open list right now, and what is the typical dollar value of an individual estimate?

YOU — Turn 19

About $310,000 in total. At our average replacement of $11,500, that works out to roughly 27 open estimates. The typical estimate is around $11,500, ranging from about $8,500 to $16,000.

VCRSoft

Tracking roughly 27 open estimates averaging $11,500 on a manual paper list that stops after seven days means thousands of dollars in replacement opportunities are dropped every month. How much of your 1.4-hour daily drive time and manual scheduling bottleneck could be reduced if your dispatcher had automated routing and geographic zone booking instead of booking in call order?

YOU — Turn 20

I can't put a number on the reduction — I've never run routed schedules to measure it against. What I know is the drive time is about 1.4 hours per tech per day now, which is about 12.6 hours a day across the 9 service techs, all because jobs are booked in call order rather than by zone. Zone booking and automated routing would cut a real chunk of that, but I don't have a measured figure for how much.

VCRSoft

Cutting drive time across 12.6 daily hours and smoothing the 3,700 annual tune-up visits currently concentrated in shoulder months directly attacks the $24,000 monthly idle labor cost and the summer capacity crunch. How are your 1,850 maintenance agreements currently logged in ServiceTitan or your office records, and are customer addresses and agreement renewal dates easily exportable for scheduling?

YOU — Turn 21

They're logged in ServiceTitan — that's where we keep dispatch, scheduling, invoicing and the agreements, with the customer addresses and renewal dates in there. The CSRs work off those dates to make about 120 renewal calls a month, about 15 minutes each, and the tune-up scheduling itself is still done by phone. I haven't tried exporting the agreement list out of ServiceTitan for scheduling, so I can't tell you how clean that export would be.

VCRSoft

Which of these improvement initiatives would you want to prioritize first for implementation within your 6-month payback and $10,000 setup budget constraints?

YOU — Turn 22

Dispatch and the maintenance agreements first — zone routing and automated scheduling, standardised truck stock, and spreading the 3,700 tune-up visits through the year including winter, plus sorting out whether the agreement price needs to go from $189 to $219. That's inside the $10,000 setup and pays back inside 6 months through the overtime, the idle shoulder time and the callbacks. Hiring two more techs comes after that, not before — I've come around on that: two more techs with the same manual dispatch just means more idle time in the shoulders.

VCRSoft

Prioritizing zone routing, standardized truck stock, and shoulder-season tune-up smoothing before adding technicians directly targets the $24,000 monthly idle labor and $18,000 summer overtime waste within your budget constraints. What specific operational steps or dispatch rules would your dispatcher need to put in place to transition from booking jobs in call order to geographic zone routing across your service area?

YOU — Turn 23

First, split the service area into zones — Austin itself, Round Rock, Cedar Park and Pflugerville — and book jobs into the zone's day rather than in the order the calls come in. Rosa, my dispatcher and office manager, would build the next day's routes in ServiceTitan each afternoon instead of working off the whiteboard and phone calls. Vans get a standard stock list so the tech has the parts for that zone's jobs before he leaves. And I'd need the senior techs bought in — they prefer phone dispatch and they'll resist routed schedules, so I'd

have Rosa trial it with one zone first and show them the drive time coming down.

VCRSoft

Trialling zone routing in ServiceTitan with Rosa while standardizing van stock and smoothing tune-up visits creates the operational foundation needed before adding technicians. What specifically does Rosa need to check or verify when building those afternoon routes in ServiceTitan to ensure emergency calls can still be inserted into a zoned day without breaking the schedule?

YOU — Turn 24

She'd need to leave a buffer in each zone — a slot or two kept open for same-day emergencies, since we promise about a 3-hour emergency response, and check which tech is first up for emergencies that day. She'd also need to see the parts on the van and how long each job should take, so one emergency doesn't blow up the whole zone's day, and make sure the after-hours answering service knows the zoned plan — we get about 35 after-hours calls a month in summer. Basically, every zoned day keeps emergency capacity in it; it doesn't get filled solid.

VCRSoft

Keeping unallocated emergency buffer slots in every zone while coordinating after-hours intake preserves your three-hour response promise without breaking scheduled density. Looking at your 1,850 maintenance agreements currently logged in ServiceTitan, how would Rosa coordinate the 3,700 annual tune-up visits so they shift into winter and early summer without overloading customer communication or requiring extra headcount?

YOU — Turn 25

Rosa would work from the ServiceTitan list by renewal month and stagger the visits across the year instead of bunching them into spring and fall — winter is at about 72% utilisation, so there's room there, and she'd batch them by zone along with the regular routing so the techs aren't criss-crossing town. The CSRs already make about 120 renewal calls a month, so they'd book the tune-ups ahead on those calls rather than a separate round of ringing. I'd approve automated reminders as a backup, though the CSRs worry they'll sound impersonal, so the calls stay. And it needs no extra headcount — that's the point: the visits move into months where the techs are sitting idle anyway.

VCRSoft

Shifting the 3,700 annual maintenance tune-ups into winter and early summer utilizes hours where techs would otherwise sit idle, while zone routing and truck stock standardisation directly attack your 1.4-hour daily drive time and 37 monthly callbacks. For the 12 monthly callbacks caused specifically by standard parts missing from the truck, what review or sign-off process would you require from your senior technicians when restocking vans to ensure compliance with your truck inventory standards without pushing back against the new routing?

YOU — Turn 26

I'd have the senior techs sign off a standard restock list each week — they check the van against the list and sign it, and Rosa checks it before the routes go out on Monday. It stays their list — they know what breaks in Austin houses — but it's the same list on every van, and it's quick, five minutes at the end of the week, not paperwork that fights the new routing.

VCRSoft

A weekly five-minute van restock sign-off enforced by Rosa gives you accountability on standard parts without creating administrative friction. For your licensed residential repair, replacement, and installation work in Austin, who verifies that local permits, municipal inspections, EPA Section 608 refrigerant recovery rules, and building or mechanical codes are met on every job, and how do you document that those safety and compliance checks are complete?

YOU — Turn 27

It's on me as the licensed owner. We pull a permit on every replacement — about $185 each, 3 to 5 days in Austin — and the techs run the safety checks, carbon monoxide and gas lines, on every furnace visit. EPA Section 608 rules apply on the refrigerant work, and the manufacturer warranty registration goes in within 30 days. The documentation is manual, though — change orders are on paper forms and get entered into ServiceTitan 2 to 4 days later, and about 15% of warranty registrations go in late, so it's not as tight as it should be.

VCRSoft

Operating under your license for compliance while Rosa handles dispatch routing and weekly van stock sign-offs means the operational changes rest entirely on internal accountability without needing external hires. Given Rosa's role managing dispatch, scheduling tune-up smoothing, and inventory sign-offs, how many hours a week can she dedicate to implementing these changes alongside her daily dispatch duties without running into capacity limits?

YOU — Turn 28

8 hours a week. Rosa is my office manager and dispatcher, and that's the time she has alongside her daily dispatch duties — she's the one who'd run the changes, and I measure the pilot weekly from her ServiceTitan reports.

VCRSoft

Dedication of eight hours a week by Rosa, measured weekly through ServiceTitan reports, provides a concrete internal owner for the pilot without risking daily dispatch capacity. What specific operational metrics from those ServiceTitan reports will you and Rosa review each week to determine whether the zone routing and tune-up smoothing pilot is successfully meeting your payback expectations?

YOU — Turn 29

Four numbers, every week from Rosa's ServiceTitan reports: first-time fix rate, from 72% up to 80%; agreement renewal rate, from 78% up to 84%, with any price test on new sales only; repair-to-replacement conversion, from 18% up to 24%; and shoulder-season tech utilisation, from 55% up to 70% as the tune-ups shift. If those four move, the payback is there.

VCRSoft

Reviewing first-time fix, agreement renewal, repair-to-replacement conversion, and shoulder-season tech utilisation weekly from ServiceTitan directly ties Rosa's 8 dedicated weekly hours to your 6-month payback and $10,000 setup budget. What compliance, privacy, contractual, regulatory, or customer-experience limits must any new process or tool respect?

YOU — Turn 30

A few hard limits. No AI-written customer safety or warranty language without my review. No automated price quotes without a tech visit. EPA Section 608 on refrigerant work, our Texas ACR license, permits on every replacement, manufacturer warranty registration within 30 days, and the safety checks — carbon monoxide and gas lines — on every furnace visit. Customer data stays private. And any automated customer outreach needs my approval first — the CSRs are right that reminders have to sound like us, not a robot.

VCRSoft

Operating under your Texas ACR license and EPA Section 608 with strict guardrails against automated pricing and impersonal outreach ensures that Rosa's eight weekly hours and the ten-thousand-dollar setup budget remain anchored to compliance and customer trust. Does this provisional ranking match your priorities, and what would you change?

YOU — Turn 31

Yes, it matches — dispatch and routing first, then the maintenance agreements and the tune-up smoothing, with hiring after that, not before. What I'd change: I'd put the open-estimate follow-up higher up, because $310,000 sitting open with 40% going stale is money we've already earned the lead for. And I'd keep Google Ads off the list until I know what the $5,500 a month actually returns in signed replacements. I walked in thinking hiring two more techs was the answer; the numbers in this conversation changed my mind on that.

VCRSoft

Thanks — I have prepared a draft of your growth strategy report. Please review it, and tell me what you would change, correct, or push back on before I finalize it.

DRAFT FEEDBACK

[Draft feedback] Thanks — this is close, and the diagnosis and the ranking are right. Please correct these before finalising:

1. The Growth Diagnosis section repeats the same sentence twice about replacement margin falling from 32% quoted to 26% realised — please delete the duplicate.

2. Change orders are not 'unbilled'. They average $640 on about 18% of installs, and about 40% of them are not billed until after the job is closed — billed late, not unbilled.

3. The three dollar value figures are not my numbers and I did not give you those percentages: $30,000 a month (15% of monthly revenue) for route optimisation, $4,500 a month (half the parts runs) for truck stock, and $7,200 a month (30% of idle labour) for smoothing. Please remove the dollar values or label them clearly as unvalidated assumptions, and keep the figures I did give: 12.6 drive hours a day across the crew, $9,000 a month in emergency parts runs, $24,000 a month in shoulder idle labour.

4. The tune-ups do not cause the $24,000 a month idle labour cost — they partly fill the shoulders. The idle cost comes from demand falling to about 190 service jobs a month and 2.6 jobs per tech per day. Please fix that wording in the smoothing section.

5. Please add the full maintenance-agreement economics: 1,850 agreements at $189 a year is about $349,650 a year; each visit costs about $97 against $94.50 of revenue per visit, plus an average repair discount of about $41 a year per agreement. Renewal is 78% and we add about 38 new agreements a month. And please record that I do not track replacement conversion for agreement members versus non-members.

6. Please add the hiring-pipeline facts behind postponing hiring: one service-tech role has been open 5 weeks with 2 candidates, neither EPA-608 certified, and only 6 of my 9 service techs are EPA 608 Universal.

7. The note in Risks & Metrics saying figures could not be verified and should be treated as unverified: the figures I supplied in the interview have been checked — please limit that warning to the illustrative estimates only, not to the numbers I gave.